

Grayson Gilbert

ACTIVE DoD TOP SECRET/SCI Clearance

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Education

University of Maryland, College Park

Master of Engineering, Robotics

May 2026

GPA 4.0

University of Maryland, College Park

Bachelor of Science, Mechanical Engineering

College Park Scholar – *Environment Technology Economy Scholar Program*

A. James Clark Legacy Scholarship Award Winner | Dean's List Fall 2021, Spring 2022

May 2022

Technical Skills & Certifications

- **CAD:** Siemens NX, Solidworks, Autodesk Fusion | **Software/Programming:** Python, C++, Linux, Bash | **Simulation/Robotics:** ROS2, MuJoCo, Gazebo, OpenCV, MATLAB, Embedded Systems | **Development Tools:** Git, Google Test, Docker, Jenkins
- **Fabrication:** CNC Routing, 3D Printing, Laser Cutting, Rapid Prototyping
- **Certifications:** FAA Part 107 Remote Pilot, FCC Amateur Radio Technician

Work Experience

Principal System Engineer

July 2026—Present

Hardware Engineer

April 2024—July 2026

Northrop Grumman, Dolphinworks Sonar Development Lab — Annapolis, MD

- 2x Bravo award recipient in 2026 for outstanding engineering and business impact
- Awarded "Top Performer" rating in 2025 (top 10% of employees at job level) for exceptional performance and contributions in engineering and business impact
- IPT Lead Engineer for next-generation multi-spectral Synthetic Aperture Sonar (SAS) sensor
- Modeled production components and tooling in Siemens NX, and produced ASME-compliant drawings for manufacturing
- Developed and performed factory acceptance testing for next generation SAS sensor
- Designed, fabricated, and integrated a man-portable SAS payload onto multiple Autonomous Underwater Vehicles (AUV) platforms; extensive on-water experience as vehicle operations and test lead
- Led SAS mechanical and software integration on customer AUV, and served as subject matter expert; traveled to customer site and integrated directly with customer teams
- Developed Python-based controller to autonomously operate payload, reducing data retrieval and analysis time by 50%
- Developed multithreaded C++ navigation interfaces utilizing serialized UDP data streams to seamlessly bridge embedded sonar hardware with AUV navigation sensors

Model Integration Lead

October 2023—March 2024

Northrop Grumman Pathways Program — Annapolis, MD

- Awarded "Top Performer" rating in 2023 for exceptional performance and contributions in mechanical engineering
- Developed 5 high fidelity MATLAB Simulink/Simscape models for a complex submersible system including hydraulics, life support, high-pressure air, and 2 ballast models.
- Established interfaces of several sub-models to ease integration into a larger system level performance model and part of a broader MBSE approach
- Served as main point of contact for all MATLAB modeling related questions on the program, and successfully resolved other teammates problems reducing down time

Hydraulics System Development Team Member

May 2023—October 2023

Northrop Grumman Pathways Program — Annapolis, MD

- Developed a high-fidelity hydraulics model for a complex submersible using MATLAB Simulink/Simscape, reducing development costs associated with physical prototyping
- Validated the model with hand calculations and real-world data from analogous systems
- Derived system requirements and leveraged the model to modify and verify the design against performance criteria

Associate Manufacturing Process Engineer

August 2022—May 2023

Northrop Grumman Pathways Program — Surface Mount Technology (SMT) – Baltimore, MD

- Led operator downtime monthly review meeting to recognize and drive SMT manufacturing process improvement
- Oversaw and approved Autoline manufacturing process as lead Duty Engineer on a weekly basis
- Developed and optimized both Stencil Print and Solder Paste Inspection programs to minimize production CCA defects

Zero-shot Sim2Real Control of Furuta Pendulum

April 2026—May 2026

Deep Reinforcement Learning Project – College Park, MD

- Achieved zero-shot sim2real transfer of a Proximal Policy Optimization (PPO) reinforcement learning policy to control an underactuated Furuta pendulum
- Trained a robust control policy within a custom Gymnasium environment featuring domain randomization, noise injection, and action latency buffers, utilizing a high-fidelity MuJoCo physics simulation
- Designed and prototyped hardware in SolidWorks, assembling 3D-printed parts with a brushless motor, magnetic encoders, and an ESP32 microcontroller
- Developed dual-core C++ embedded software for the resource-limited ESP32 to evaluate the policy and execute precise Torque Field Oriented Control (FOC)
- Validated system robustness through physical endurance testing, successfully maintaining an active, continuous balance for over an hour

Multi-Agent Robotic SLAM Simulation

November 2025—December 2025

Autonomous Mobile Robot SLAM Project – College Park, MD

- Designed a multi-agent SLAM architecture in ROS 2 and Webots to coordinate autonomous robot fleet area mapping
- Developed a C++ map-fusion node to mesh distributed occupancy grids from independent robots into a single global map
- Engineered sector-based exploration algorithms for waypoint navigation to survey unknown indoor environments
- Constructed CI/CD pipelines using GitHub Actions, enforcing TDD via GoogleTest to ensure high code reliability

Monocular Perception and Tracking Library

September 2025—November 2025

OpenCV, DepthAnything, and YOLOv5 Project – College Park, MD

- Engineered a C++ perception module fusing YOLOv5 object detection and Depth Anything models to accurately detect, track, and estimate the 3D position of humans from monocular video
- Developed an object-oriented C++ library using OpenCV 4.10, providing an adaptable framework for implementing advanced monocular depth estimation and tracking in custom robotics applications
- Implemented a robust CI/CD pipeline using GitHub Actions and CMake, TDD and ensuring high code quality across the perception stack
- Architected and deployed the module using Docker, incorporating comprehensive code coverage analysis and automated Doxygen documentation generation

Autonomous Barista Robot

November 2024—December 2024

6-DOF Robot Manipulator Project – College Park, MD

- Developed a ROS2 package to enable autonomous operation of a 6-DOF Universal Robots UR10e manipulator
- Simulated autonomous coffee-making tasks in a custom Gazebo environment, utilizing RViz for sensor data visualization
- Designed and implemented position, velocity, and effort controllers for precise robotic manipulation in a physics-based simulation.
- Applied inverse velocity kinematics to compute joint positions, ensuring accurate tracking of the desired end-effector trajectory

Extended Kalman Filter (EKF) based Linear Quadratic Regulator (LQR)

September 2024—November 2024

Underwater Optical Wireless Communication (UOWC) Control Project – College Park, MD

- Re-derived and validated results from an IEEE journal paper implementing an Extended Kalman Filter (EKF) algorithm and a Linear Quadratic Regulator (LQR) controller for optimal stabilization of an UOWC system
- Simulated system dynamics and control performance using MATLAB and Simulink
- Developed state-space representations and conducted system analysis for the communication model
- Optimally calculated Kalman and LQR gains to enhance system stability and control performance